



Mahendra Akula

Senior SDET & Automation Architect | Playwright · Cypress · CI/CD · Cloud | AWS Certified

mahendrasakula@gmail.com | 609-874-4195 | Dallas–Fort Worth, TX |
linkedin.com/in/mahendraakula
Visa: H1B | I-140 Approved



PROFESSIONAL SUMMARY

Senior SDET and Automation Architect with 13+ years of experience building enterprise-grade automation frameworks, scalable CI/CD pipelines, and cloud-based QA platforms across FinTech, SaaS, Healthcare, E-commerce, and Public Sector domains. Deep hands-on expertise in Playwright (cross-browser, API, component, and accessibility testing), Cypress, REST Assured, JUnit, TestNG, and Maven. Strong DevOps background with Jenkins, GitHub/Bitbucket, Azure DevOps, and AWS. AWS Certified AI Practitioner and AWS Certified Generative AI Developer – Professional (Early Adopter, Top 5,000 globally).

AWS CERTIFICATIONS

AWS Certified Generative AI Developer – Professional

Early Adopter — Among the first 5,000 worldwide | Amazon Bedrock · Foundation Models · RAG Pipelines · Prompt Engineering · AI Agents · LLM Evaluation

AWS Certified AI Practitioner

Foundational | AI/ML on AWS · Machine Learning · Cloud AI Services

PROFESSIONAL EXPERIENCE

Senior Software Development Engineer in Test

May 2023 – Present

Compunnel Inc (Client: Finra) | New Jersey, United States · Remote

- Architect and lead Playwright-based automation framework covering cross-browser UI, API, component, and accessibility testing across regulatory financial applications.
- Design multi-layer test strategy spanning unit, integration, functional, and scenario testing using Playwright, Cypress, REST Assured, and Postman; significantly improving regression stability and release confidence.
- Engineer and optimize CI/CD pipelines in Jenkins (monorepo) and Azure DevOps — including branching strategies, merge gates, version control workflows, and deployment automation with shift-left quality enforcement.
- Develop SQL queries and implement data seeding strategies to support automated test scenarios across complex financial data environments.
- Lead performance testing and contribute to chaos testing and security testing activities to validate system resilience.
- Use Amazon Q, Kiro CLI, and LLM tools to accelerate test case authoring, generate edge-case scenarios, and automate failure triage via Playwright MCP integrations.
- Apply AWS Generative AI and Amazon Bedrock to build intelligent test generation pipelines; actively contribute to cloud environment management on AWS.
- Evaluate and recommend emerging testing tools and frameworks; led adoption of Playwright replacing legacy WebDriver-based suites.

Senior Software Development Engineer in Test

April 2022 – May 2023

Friendly Consultants (Clients: Fannie Mae, Northern Trust, Best Buy) | New Jersey, United States

- Designed scalable Playwright and Cypress automation frameworks (SpecFlow/Gherkin + JavaScript) from scratch for multifamily loan workflows at Fannie Mae, covering functional and integration layers.
- Built REST API automation using REST Assured/RestSharp and Java; validated complex back-end workflows, ETL batch processes, and database interactions with custom SQL data seeding strategies.
- Architected Playwright and BDD-based test suites for wealth-management portals at Northern Trust, including accessibility and cross-browser coverage.
- Developed UI and API automation (Java + Selenium + REST tools) for Best Buy's omni-channel platform; maintained build/release CI/CD definitions in Azure DevOps with Git-based branching strategies.
- Contributed to cloud migration efforts on AWS and supported ongoing cloud environment management and observability improvements.

Senior Software Engineer in Test

February 2020 – April 2022

Freshworks (Freshservice ITSM) | India

- Built hybrid frameworks (Keyword + Data-Driven) for backend and REST API automation using REST Assured, Ruby, RSpec, and Selenium; introduced Playwright for front-end regression automation.
- Designed and maintained CI/CD pipelines (Jenkins, Bamboo) with automated test gates, version control policies, and deployment automation in monorepo environments.
- Worked with AWS clusters, Kubernetes, and Docker as part of build and deployment pipelines; supported performance testing with load scenario design.
- Implemented SQL-based data seeding and validation strategies to support automated testing of ITSM workflows and API integrations.

Senior QA Analyst / SDET

July 2018 – February 2020

ValueLabs (Client: WebMD/Medscape) | India

- Implemented BDD frameworks using Cucumber + Selenium POM for healthcare content and services; evaluated and introduced Playwright for improved cross-browser and accessibility testing coverage.
- Automated UI and API flows using Selenium WebDriver, TestNG, Postman, and Swagger; performed microservices and integration testing.
- Developed SQL queries for test data management and contributed to test coverage analysis and quality reporting.

Senior Software Test Engineer

February 2016 – July 2018

Tech Mahindra (Client: NH DMV – MOVES) | India

- Led end-to-end testing for DMV modernization (vehicle registration, licensing, financial responsibility modules) across functional, integration, and regression layers.
- Designed test cases using Conformiq Creator (MBT) and Selenium WebDriver + TestNG; built CI pipelines using Jenkins for automated regression execution.

Senior Software Engineer / SDET

August 2012 – February 2016

Ideabytes Inc | India

- Developed and executed automation frameworks using Selenium WebDriver, TestNG, SoapUI, and SQL across multiple customer projects spanning web and API layers.

CORE SKILLS

Playwright & Web Automation: Playwright (cross-browser, API, component, accessibility), Cypress, Selenium WebDriver, Selenium Grid, BrowserStack, Perfecto, Galen Framework

API, Integration & ETL Testing: REST Assured, Postman, RestSharp, SoapUI, JUnit, TestNG, Maven, ETL batch testing, microservices validation, SQL data seeding

CI/CD & DevOps: Jenkins Core (monorepo, LWC), Azure DevOps, GitHub Actions, Bamboo, Maven; Git, GitHub, Bitbucket/Stash; branching strategies, deployment automation

Cloud & Infrastructure: AWS (S3, Bedrock, EC2, AI Services), Azure; Docker, Kubernetes; cloud migration, environment management

AI & Generative AI: Amazon Bedrock, Foundation Models, RAG Pipelines, Prompt Engineering, AI Agents, LLM Evaluation, Amazon Q, Kiro CLI, MCP Integrations

Performance & Security Testing: Performance testing, chaos engineering, security testing support, load scenario design, BrowserMob Proxy

Observability & Monitoring: Splunk, Datadog, Dynatrace, New Relic; production system health monitoring, failure analysis

Programming: Java, JavaScript, Node.js, Ruby, SQL

Test Frameworks: Hybrid, POM, Data-Driven, Keyword-Driven, BDD/ATDD (Cucumber, SpecFlow, Gherkin), Serenity, MBT, accessibility standards (WCAG)

Quality & Process: STLC, Agile/Scrum, Waterfall, V-Model; Test Strategy, Test Plans, RTM, Impact Analysis; Jira, Azure DevOps, Rally

EDUCATION

Master's Degree, Computer Programming, Specific Applications

June 2008 – August 2011

Osmania University, Hyderabad, India

Bachelor's Degree, Mathematics and Computer Science

June 2005 – March 2008

Kakatiya University, Warangal, India

Higher Secondary Certificate

June 2003 – March 2005

Board of Intermediate Education, Andhra Pradesh, India

VISA STATUS

H1B Visa | I-140 Approved (EB Category) — Authorized to work in the United States